FPT Education

FPT UNIVERSITY

DBI202_IA2001

PROJECT

Group 2

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Team Members

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1.PROJECT INTRODUCTION

System Name: Cinema Booking System

Problem Statement:

Manual cinema management can lead to:

- Ticketing errors
- Difficulty in seat control
- Lack of revenue tracking

System Objectives:

- Automate the ticket booking process
- Centralize data management
- Support decision-making through analytics

2.SYSTEM SCOPE

The system serves two main user roles:

Staff :

- Login
- Sell tickets
- Select seats
- Add food and beverages

2. SYSTEM SCOPE

Manager

- Manage movies
- Manage showtimes
- Manage staff
- View reports and statistics

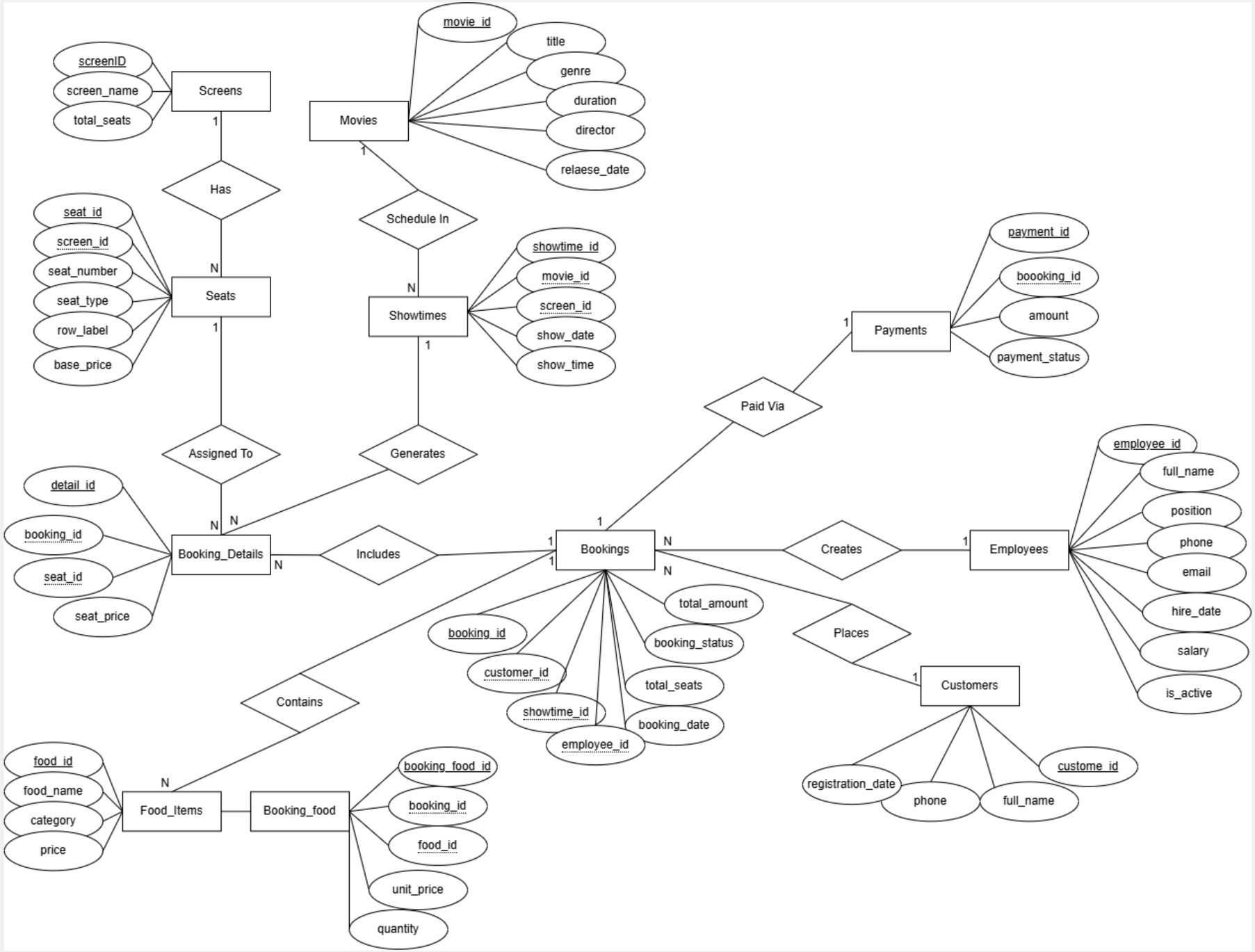
Screens
screenID
screen_name
total_seats
Has
Seats
seat_id
screen_id
seat_number
seat_type
row_label
base_price
Assigned To
Booking_Details
detail_id
booking_id
seat_id
seat_price
Movies
movie_id
title
genre
duration
director
relaese_date
Schedule In
Showtimes
showtime_id
movie_id
screen_id
show_date
show_time
Generates
Includes
Bookings
booking_id
customer_id
showtime_id
employee_id
total_amount
booking_status
total_seats
booking_date
Contains
Food_Items
food_id
food_name
category
price
Booking_food
booking_food_id
booking_id
food_id
unit_price
quantity
Paid Via
Payments
payment_id
boooking_id
amount
payment_status
Creates
Employees
employee_id
full_name
position
phone
email
hire_date
salary
is_active
Places
Customers
custome_id
full_name
phone
registration_date

Table	Primary Key (PK)	Foreign Key (FK)	Description
Movies	movie_id	—	Stores movie information: title, genre, duration, rating, director, release date.
Screens	screen_id	—	Cinema screen information: screen name, total seats.
Seats	seat_id	screen_id	Seats within a screen: seat number, row, seat type (Standard/VIP/Couple), base price.
Showtimes	showtime_id	movie_id, screen_id	Showtime schedule: which movie plays in which screen, date/time, and base ticket price.
Employees	employee_id	—	Staff records: full name, position, phone, email, hire date, salary, password, active status.
Customers	customer_id	—	Customer records: full name, phone number (unique).
Bookings	booking_id	customer_id, showtime_id, employee_id	Booking orders: customer books via an employee, booking date, order status.
Booking_Details	booking_detail_id	booking_id, seat_id	Booking line items: each row represents one seat in one booking, price at time of booking.
Food_Items	food_id	—	Food & drink menu: item name, price.
Booking_Food	booking_food_id	booking_id, food_id	Food items attached to a booking: quantity and unit price at time of purchase.
Payments	payment_id	booking_id	Payment records: which booking was paid, method (Cash/Card...), timestamp, and amount.

3. MAIN FEATURES

3.1. Authentication & Security

Login using:

- Employee ID
- Password

System validation:

- Account existence
- Correct password
- Active status

Additional feature:

- Change password

3.2. Movie Management

Display movie list:

- Title, genre, duration
- Director, rating

Functions:

- Add new movie
- Delete movie (with constraints)

Constraint:

- Movies cannot be deleted if tickets have already been booked

3.3. Showtime Management

Create showtimes including:

- Movie
- Screening room
- Date & time
- Ticket price

Display:

- Number of booked / available seats
- Status (available / sold out)

3.4. Seat Management & Booking

Display seat layout by theater room:

- Standard seats
- VIP seats
- Couple seats

Seat Status:

- Available
- Booked
- Selected

Rules:

- Maximum 10 seats per booking
- Cannot select already booked seats

3.5. Food & Combo Ordering

Popcorn:

- Flavors: Caramel / Cheese / Mix
- Sizes: S, M, L

Drinks:

- Pepsi, Coca, 7Up, Fanta

Combos:

- Popcorn + Drink

  → Optional (can be skipped)

3.6. Order Confirmation & Payment

Display Information:

- Movie details
- Showtime
- Selected seats
- Food & drinks
- Total price

Customer Information:

- Full name
- Phone number

Validation:

- Must be 10 digits
- Must start with 0
- Numeric only (no letters)

3.8. Reports & Analytics

Revenue:

- Monthly / Yearly

Statistics:

- Staff ticket sales
- Number of tickets / orders

Customer Data:

- Booking history

Display:

- Data tables with clear visualization

3.8. Staff Management

- View staff list
- Add new staff

Role Assignment:

- Staff
- Manager

4.SYSTEM ARCHITECTURE

SYSTEM ARCHITECTURE

1. Frontend :

- Tkinter (Python)
- Step-by-step UI (5 steps)

2. Backend:

- Logic
- Calculation
- Validation

3. Database

- SQL Server
- Tables, Procedures, Functions

WORKFLOW

1. Login
2. Select movie
3. Select showtime
4. Select seats
5. Add food
6. Enter info
7. Payment

STRENGTHS

- User-friendly UI
- Clear roles
- Strong validation
- Integrated system
- Analytics support

LIMITATIONS

- No online payment
- No web/mobile version
- Local deployment only

FUTURE DEVELOPMENT

- Web/App (React, Flutter)
- Online payment integration
- AI movie recommendation
- Loyalty system
- Online booking

CONCLUSION

- Efficient cinema management
- Reduce manual errors
- Scalable system

Thanks you for your listening !